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Valuing energy flexibility from water systems

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Water systems represent an untapped source of electric power load flexibility, but determining the value of this flexibility requires quantitative comparisons to other grid-scale energy storage technologies and a compelling economic case for water system operators. Here we present a unified framework for representing water asset flexibility using grid-scale energy storage metrics (round-trip efficiency, energy capacity and power capacity) and assessing the technoeconomic benefits of energy flexibility at the water facility scale (levelized cost of water and levelized value of flexibility). We apply this framework to case studies of an advanced water treatment (desalination) plant, a water distribution network and a wastewater treatment plant. The framework reveals strengths and limitations of water system flexibility relative to other grid-scale energy storage solutions, high-value opportunities for flexible load operation of water assets and the critical role of electricity tariff structures and energy service markets in determining water sector participation in load flexibility. Ultimately, this unified framework for valuating water asset flexibility enables both electricity and water asset managers to prioritize investments based on levelized cost comparisons across their respective portfolios.

Most pathways for decarbonizing the electricity sector by 2050 include substantial increases in intermittent renewable generation paired with several TWh of grid-scale energy storage[1–3]. Grid-scale energy storage refers to technologies with the ability to shift the net demand for electricity over time[4]. Options for grid-scale storage technologies broadly include chemical or thermal storage (for example, molten salt, phase change materials), bidirectional electricity storage (for example, lithium-ion, redox flow) and flexible generation or controllable loads (for example, electric vehicles, building systems, water infrastructure and so on)[5].

The value and performance of these diverse sources of grid-scale electricity storage are frequently described in terms of their round-trip efficiency (energy returned after storage), energy capacity (amount of energy that may be stored or shifted), power capacity (the speed at which energy can be charged or discharged) and cost (the cost to install and operate a device over its lifetime)[6]. Other critical metrics include: dispatchability, durability and stability. Grid operators and investors use these metrics to compare diverse technologies, determine the net present value of marginal energy storage and minimize the costs of grid stabilization.

Whereas these metrics are readily derived for stand-alone storage technologies (that is, chemical, thermal, bidirectional), we are unaware of a unified framework for characterizing the value of flexible load energy services (that is, virtual batteries) across diverse, multifunctional assets. The closest equivalent is prior work developing analogues between stand-alone storage and flexible thermostatic loads from buildings[7–10]. A generalizable approach for valuing energy flexibility assets across diverse sectors would assist electric power grid operators in assessing the dispatchability of flexible loads, the potential for demand response to mitigate resource intermittency and the lowest cost pathways for reliable power supply[11,12]. A generalizable framework would also support sector-specific efforts to assess the extent and value of load flexibility across asset classes and individual facilities. Finally, asset operators could use this framework to maximize the net present value of flexible load operation by managing their assets to minimize electricity bills or generate revenue through participation in energy services markets.

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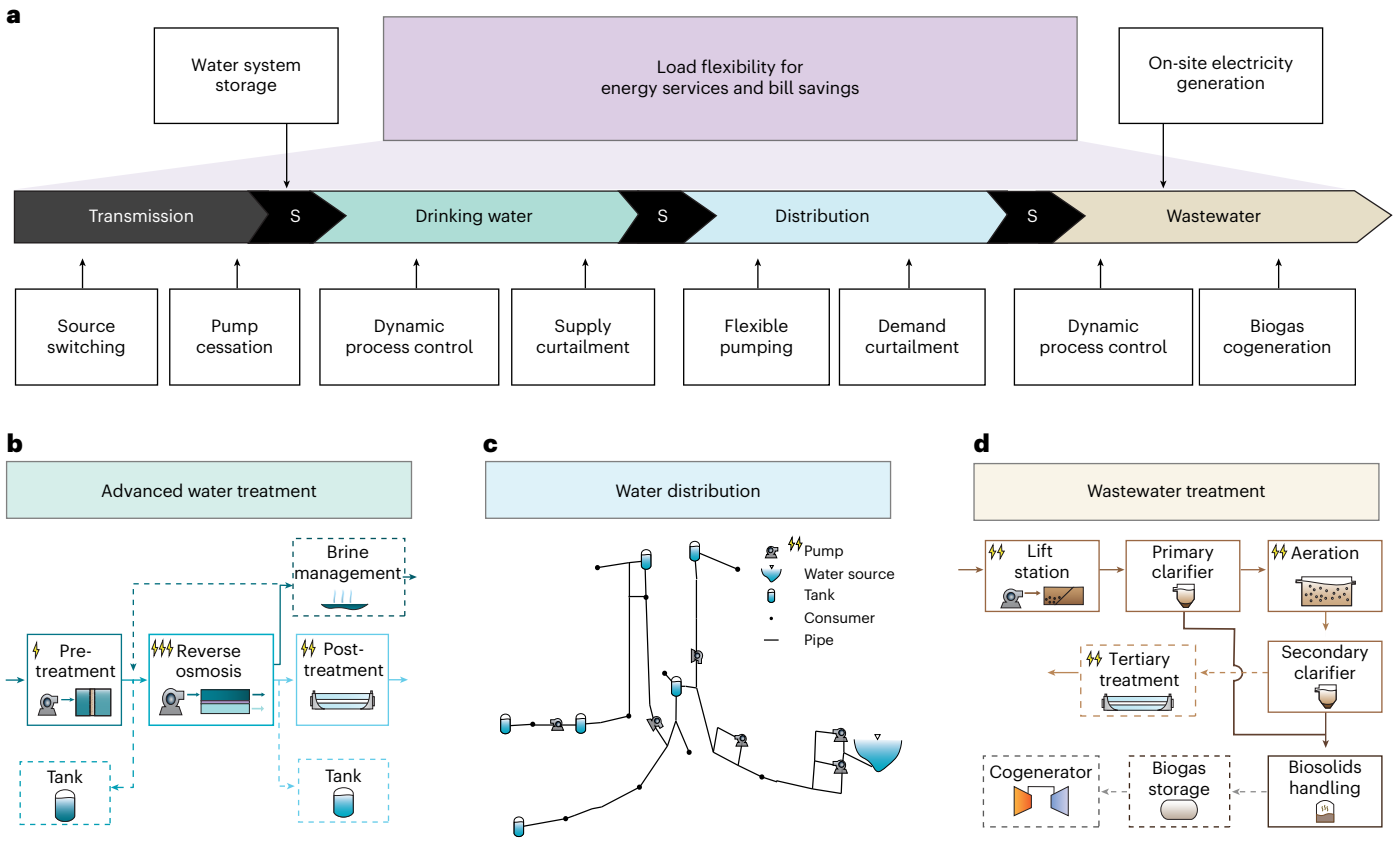

Fig. 1 | Mechanisms of load flexibility for energy services and bill savings in the water sector. a, Mechanisms for load flexibility depicted for each stage of the water supply chain. **b**–**d**, Canonical systems diagrams of the case studies described in this study are depicted in advanced water treatment (**b**), water distribution (**c**) and wastewater treatment (**d**). 'S' signifies water storage capacity via tanks or natural reservoirs. Lightning bolts within each unit process qualitatively denote relative energy intensity. Shading relates to the concentrations of salt or waste (darker refers to higher concentration). Dotted lines refer to features that are not universal.

The need for a generalizable framework is especially acute for valuing flexibility in critical infrastructure assets. Here load flexibility can introduce tension between domain-specific management priorities (for example, product quality, product value, energy service capacity) and cost savings or revenue generation. For example, in the water sector, a hierarchy of management priorities is codified through regulation. Human safety, water quality and system reliability[13] take precedence over minimizing operating costs, generating new revenue streams or minimizing carbon emissions. Providing energy services (for example, demand response, frequency regulation and so on) may advance the worthy goal of enhancing system affordability, but the degree of energy flexibility in the water asset, and the extent to which flexibility is deployed, depend on first meeting water system reliability targets.

Assessing and deploying latent flexibility in critical infrastructure assets also requires a generalizable approach for evaluating that flexibility across asset classes and individual facilities. Within the water sector, the differences in the latent flexibility of drinking water treatment, water distribution and wastewater treatment assets are a result of underlying differences in process dynamics, supply chain constraints and asset coupling, the degree of digitization and economic structures. For example, each facility is a network of unit processes with different start-up/shutdown attributes and process dynamics. Coupling within facilities and across assets (for example, treatment plants, reservoirs and so on) can constrain the operation of upstream and downstream operations. The physical and digital systems for start-up, shutdown or modifications to system operation are not always controllable on the sub-hourly timescales that determine electricity prices[14]. And the net financial benefits of load flexibility are highly dependent on the tariff structures and energy services markets in which the facility is operating[15,16]. The large variance of these factors across facilities and asset classes has impeded a generalized understanding of the potential for, and value of, load flexibility at water facilities.

Despite this variance, the water sector shares key features of other sectors with high-value flexible loads: large aggregate energy consumption, high energy costs and capacity for real-time control[11]. Water and wastewater utilities account for approximately 4% of global electricity demand (0.6–6.2% on a regional basis), though it is uncertain what fraction of this load is flexible[17,18]. Energy intensity and electricity costs vary considerably across the water sector[19,20]; electricity typically represents approximately 30–50% of advanced water treatment lifetime costs, 60–80% of distribution and transmission costs and 20–30% of wastewater treatment costs[20–22]. Finally, the water sector's capacity for real-time control has expanded dramatically in recent years, though legacy systems lack the industrial automation equipment required for flexible load operation[23].

We believe that both water asset managers and electric power grid operators would benefit from a unified, quantitative framework for valuing the net benefits of energy flexibility from water assets and comparing that value to other grid-scale energy storage solutions. Any such framework must explicitly account for increased compliance risk for water infrastructure and accommodate diverse operating scenarios and flexibility mechanisms with nonlinear dynamics. It must be scalable to accommodate different facility designs, capacities, network structures and data ingestion and representation approaches. It must accommodate a variety of economic data such as electricity

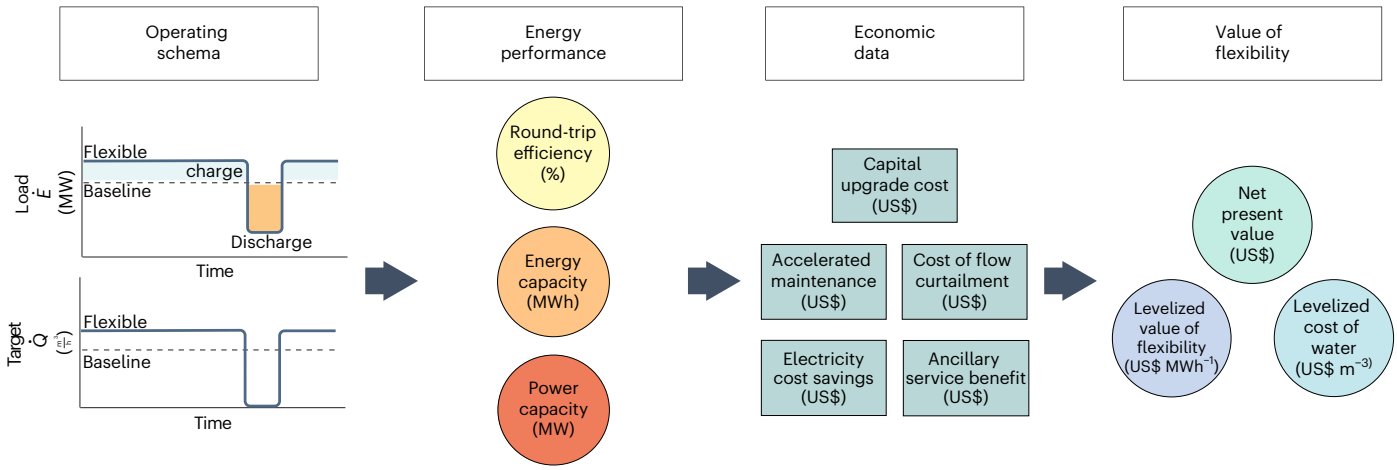

Fig. 2 | Information flow. The proposed approach to characterize and value flexible operating schema in terms of a baseline operating schedule and economic data. $\dot{E}$ denotes the power consumption, or electrical load, of the facility; and $\dot{Q}$ denotes the target flow rate (water treatment, water production or water delivery) associated with the given system.

tariff structures, capital upgrades and depreciating assets. Finally, the framework must align with existing decision-making frameworks that are used by both grid and water system operators.

This paper begins by reviewing the forms of load flexibility that are available to water systems, with a particular focus on desalination (advanced water treatment), water distribution and wastewater treatment. Next, we introduce a set of equations for calculating the round-trip efficiency, energy capacity, power capacity, the net present value of flexibility and the levelized cost of flexibility in water facilities. These metrics may be applied to any water facility, irrespective of the scale of the facility, original facility design, downstream supply chain constraints or resolution of measurements. We then demonstrate the calculation of these metrics using simplified energy models of a seawater desalination plant, a water distribution network and a wastewater treatment plant and using an existing tariff structure from Santa Barbara Clean Energy. We also perform scenario analysis using existing electricity rates from water utilities in California, Texas, Florida and New York to explore the effects of electricity rate structure design on the value of flexibility. Finally, we discuss the implications of our findings on the growth potential for load flexibility resources from the water sector.

Mechanisms for energy flexibility in water assets

The water sector has a diverse set of mechanisms for delivering energy flexibility (Fig. 1a). In water transmission infrastructure, operators can modulate energy load by switching between different raw water sources with different energy intensities or by temporarily turning off pumps and relying on stored water[22]. In drinking water infrastructure (Fig. 1b), load can be dynamically controlled by modulating the water recovery or treatment intensity[13,24,25]. Supply curtailment can also be used to temporarily reduce, within safety limits, the amount of water that is sent to the distribution network or storage[26,27]. The water distribution system can leverage extensive water storage (Fig. 1c) to minimize short-term pumping load on intermittently operated pumps[28–31]. Demand curtailment, or efforts to temporarily reduce water consumption through price signals or explicit conservation mandates, are typically used during droughts but could conceivably be used over shorter time durations[32]. Finally, wastewater systems (Fig. 1d) can use dynamic process control to vary volume of treatment flow, aeration and biogas production over time[33–35]. Energy flexibility is further enhanced in facilities with on-site biogas generation and cogeneration. The ability to compare the costs and benefits of these diverse flexibility mechanisms across the water system will be essential to their adoption by water system operators.

Energy flexibility performance and cost metrics

We propose a framework and associated set of equations for characterizing water system load flexibility using metrics for energy performance that are directly analogous to other grid-scale energy services (Fig. 2). As highlighted in the introduction, key metrics of merit for grid operators include round-trip efficiency, energy capacity (MWh) and power capacity (MW). When combined with economic parameters (for example, accelerated maintenance, electricity cost savings and so on), this framework provides a generalizable approach for valuing this load flexibility in terms familiar to water system operators: net present value, the levelized cost of water and the levelized value of flexibility.

Round-trip efficiency

Round-trip efficiency (RTE) (η_{RT}, %) or energy storage efficiency (equation (1)) is defined as the ratio between the power consumed during the discharging (E_{out}) and charging (E_{in}) phases of operation[8,12].

$$\eta_{RT} = \frac{E_{out}}{E_{in}} = \frac{\int^{D} \dot{E}_f \, dt}{\int^{C} \dot{E}_f \, dt} = \frac{\int^{T} \max(0, \dot{E}_0 - \dot{E}_f)}{\int^{T} \max(0, \dot{E}_f - \dot{E}_0)} \quad (1)$$

Here $\dot{E}$ represents the load or power demand from the system, subscripts 0 and f denote the baseline and flexibly operated cases, respectively, and time is represented by t. The time step should be selected to match the rate structure defined in relevant energy tariffs but will most commonly be either 15 min or 1 h. The set D represents all time steps at which the flexibly operating system is effectively 'discharging' or consuming less energy than the baseline. Conversely, the set C contains the 'charging' time steps and represents the system consuming more than the baseline load. RTE is a dimensionless ratio of energy consumption and is bounded between 0 to 1.

In the case where the target flow rate ($\dot{Q}$) is not equal between flexible and baseline operating schema, this definition of efficiency does not hold, as the system does not complete a round trip. Examples of cases where a round trip may not be completed include water supply curtailment (for example, a desalination plant shutting down and deferring production to an alternative supply) and water demand curtailment (for example, water conservation efforts that reduce the volume of water pumped via a distribution network).

Energy capacity

Energy capacity (E) represents the amount of energy that can be flexibly shifted and has units of kWh or MWh. The normalized energy capacity represents the fraction of load that is shiftable.

$$E = \Delta E_D = \int^T \max(0, \dot{E}_0 - \dot{E}_f)\, dt = \int^D (\dot{E}_0 - \dot{E}_f)\, dt \quad (2)$$

$$\frac{\Delta E_D}{\bar{E}_0} = \frac{\int^D (\dot{E}_0 - \dot{E}_f)\, dt}{\int^T \dot{E}_0 dt} \quad (3)$$

The energy discharged, equation (2), represents the difference between the flexible and baseline load during discharging time steps. This can be written as an integral over the entire time horizon or the set of time steps at which the system is discharging (D). The results are normalized to baseline energy consumption over the time horizon in equation (3). It should be noted that the charging and discharging energy are related via the round-trip efficiency.

Power capacity

Power capacity (P) represents the average rate at which the system can charge or discharge and has units of kilowatts or megawatts. The normalized power capacity represents the average rate at which a load can be curtailed as a fraction of its peak power.

$$P = \frac{\int^C (\dot{E}_f - \dot{E}_0)\, dt + \int^D (\dot{E}_0 - \dot{E}_f)\, dt}{\int^T dt} \quad (4)$$

$$P_D = \frac{\int^D (\dot{E}_0 - \dot{E}_f)\, dt}{\int^D dt} = \frac{\Delta E_D}{t_D} \quad (5)$$

$$P_C = \frac{\int^C (\dot{E}_f - \dot{E}_0)\, dt}{\int^C dt} = \frac{\Delta E_c}{t_C} \quad (6)$$

$$\frac{P}{\dot{E}_{0,\max}} = \frac{\Delta E_C + \Delta E_D}{\max\limits_{t \in T}(\dot{E}_0) \int^T dt} \quad (7)$$

The power capacity can be calculated during the charging time frame (equation (4)), the discharging time frame (equation (5)) or as an average over the entire time horizon (equation (6)). The total power capacity is a time-weighted average of charging (C) and discharging (D) power capacities. The charging and discharging forms may be useful in specific contexts. For example, demand-response markets incentive facilities to shed load below their baseline rate, and the magnitude of that shed capacity is the discharging power capacity (P_D).

In this study, we analyse the dimensionless form of the total average power by normalizing by the maximum baseline power consumption, (equation (7)). It should be noted that these metrics are defined to be context specific and evaluate the relative performance between baseline and flexible operating schedules at a single facility.

Value of flexibility

The net present value (NPV) or value of flexibility is a balance between long-term returns (benefit minus costs) and upfront costs (capital expenditures).

$$\text{Costs} = \Delta\text{OM} + \text{LCOW}_{\text{baseline}} \int^T (\dot{Q}_0 - \dot{Q}_f) dt \quad (8)$$

$$\text{Benefits} = \int^T f_e(\dot{E}_0) - f_e(\dot{E}_f)\, dt + \int^T f_s(\dot{E}_f)\, dt \quad (9)$$

$$\text{NPV} = (\text{benefits} - \text{costs}) \left(\sum_{i=0}^{N} \frac{1}{(1+r)^i} \right) - \text{CapEx}_f \quad (10)$$

$$\text{ROI} = \frac{\text{NPV}}{\text{CapEx}_f} \quad (11)$$

Costs (equation (8)) account for both additional maintenance costs (ΔOM) of flexible operations and the value of lost water production associated with any flow curtailment. The additional maintenance (ΔOM) term is defined as the increase in operating and maintenance costs beyond the baseline schedule due to unique operations or accelerated component degradation. This value is zero for assets with native flexibility. The cost of flow curtailment is approximated using the levelized cost of water of the baseline ($\text{LCOW}_{\text{baseline}}$). Whereas this term equals zero when water production normally occurring during peak hours is fully shifted to off-peak hours (no net water production curtailment), in water systems with diverse supplies, it may be more profitable to curtail net production from high marginal cost plants. In this study, we introduce an intermittent advanced water treatment case that simulates curtailment of 5 h of lost baseline production (21%), which is consistent with the duration of demand-response events[14].

Benefits (equation (9)) include electricity cost savings and ancillary service payouts. The electricity cost savings stem from optimizing power consumption ($\dot{E}$) relative to baseline operation to avoid periods with high electricity rates. f_e is a function that calculates the electricity cost associated with a given load profile and tariff structure[36]. Ancillary service payouts incorporate the expected revenue from participation in programmes or markets such as demand response or frequency regulation[37]. f_s represents the function that maps the flexible load profile to the expected payout in a specified programme. Given the stochastic nature of these ancillary service markets, this study sets the ancillary service payout term to zero. For reference, the authors are aware of several instances in which ancillary service contracts with water treatment facilities are returning over 10% of the annual electricity bills in exchange for commitments to six shutdowns per year.

The net present value of flexible load operation (equation (10)) is calculated by summing the discounted costs and benefits with any upfront capital costs. The discount rate (r) is formulated and applied over N discrete interest periods. Whereas the resolution (usually annual) is not necessarily the same as the simulation time step, the overall absolute time horizon of the interest period and simulation must match. For example, 3% annual depreciation over 30 years would reduce the expected benefit by nearly 35%. Net capital costs of the flexibility investment (CapEx_f) include facility upgrades required to shift load (for example, variable speed drives, intermediate tank storage, control software) and any revenue associated with selling obsolete equipment during a flexibility retrofit.

Finally, the return on investment (ROI) for flexible load operation upgrades (equation (11)) is defined by the net present value of the investment divided by the upfront capital costs. ROI (%) is a normalized metric that aids financial comparisons independent of the scale of the water facility.

The flexibility investments are simulated for a representative year within a 30-year lifetime. The value of flexibility metric builds on prior work that assumed a fixed cost of water and solved for the net present value of flexibility at the grid scale[11]. The formulation in this paper accounts for any costs associated with flexibly operating facilities and applies facility-centric, physics-informed models that constrain flexibility to reasonable bounds (for example, target flow-rate requirements, constraints associated with process dynamics).

Levelized cost and value

The levelized value of flexibility (LVOF), equation (12), represents the net present value of flexibility to the facility per unit of energy

discharged, where both NPV and ΔE_D are calculated over the facility lifetime.

$$\mathrm{LVOF} = \frac{\mathrm{NPV}}{\int^{D} (\dot{E}_0 - \dot{E}_f)\, dt} = \frac{\mathrm{NPV}}{\Delta E_D} \tag{12}$$

$$\mathrm{LVOF} = -\mathrm{LCOF} \tag{13}$$

$$\mathrm{LCOW}_{\mathrm{flexible}} = \mathrm{LCOW}_{\mathrm{baseline}} - \frac{\mathrm{NPV}}{\int^{T} \dot{Q}_f\, dt} \tag{14}$$

In market equilibrium, the LVOF is equal to the negative of the levelized cost of flexibility (LCOF) (equation (13)). The impact of flexible load operation on the water treatment asset's levelized cost of water (LCOW) delivery is determined by subtracting the net present value of flexibility from the baseline LCOW (equation (14)).

LCOF is directly comparable to the value of other grid-scale storage options on a US$ MWh^{-1} basis. Similarly, $\mathrm{LCOW}_{\mathrm{flexible}}$ enables facile comparison of flexibility investments across the water sector and the impact of flexibility on the levelized cost of water for a specific facility. Finally, an alternative formulation of equation (12) could normalize the NPV by the power capacity (P) to get a levelized cost in units of US$ MW^{-1}.

Operationalizing energy flexibility performance metrics

Whereas baseline operations are optimized to maximize the energy efficiency of the facilities, flexible operations minimize the energy costs of the facility. The peak energy price from the Santa Barbara Clean Energy tariff case is over four times the off-peak energy price (Fig. 5), providing a strong incentive for water asset managers to flexibly operate facilities (Fig. 3a–c). In the continuous desalination case (advanced water treatment, Fig. 3a), the plant shifts load by reducing water recovery during the peak hours and increasing recovery during off-peak hours to meet daily production goals. In the intermittent desalination case (with curtailment), the system halts water production during the peak hours. In the water distribution case (Fig. 3b), water delivery is moved earlier in the day (hours 3–10) to allow the pumps to shut down during the peak hours while still meeting water delivery requirements. In the wastewater treatment case (Fig. 3c), the influent storage tank is allowed to fill during the peak hours, enabling the system to reschedule energy-intensive treatment steps to off-peak hours without violating treatment constraints or overfilling intermediate tanks. Each system exhibits a non-sheddable load or a minimum amount of energy that is required to sustain operations and meet system-specific target flow constraints (that is, daily water treatment minimums, hourly water delivery requirements, sub-hourly influent treatment demands).

The technical and economic conditions in the case studies (Fig. 3d–i), generally result in optimized load flexibility with high round-trip efficiency (greater than 85%) but relatively low energy and power capacities (less than 30%). For reference, the round-trip efficiency of the baseline lithium-ion battery is 86%, whereas the energy and power capacities depend on the battery specifications[6]. Systems that are right sized or have tight operational constraints often have low energy and power capacities. Critical water treatment, distribution and wastewater infrastructure is typically both right sized (to minimize costs) and tightly operationally constrained (to maximize safety and reliability). The desalination case with curtailment generally has the highest energy and power capacity, but an undefined round-trip efficiency (see note in definitions section) because the water production target is reduced. As shown by the continuous case of desalination, simply using process control (for example, variable water recovery) as a flexibility mechanism results in approximately three times less energy and power capacity than the intermittent mode of flexibility (Fig. 3d).

Normalized capacities (Fig. 3d–i) are adjusted by the respective baseline load to obtain the metrics in absolute units (MWh or MW). For example, a desalination system operating at a constant baseline of 1.8 MW (43 MWh per day) with normalized energy and power capacities of 10% and 20%, respectively, would have an absolute energy and power capacity of 4.3 MWh and 0.9 MW.

The electricity tariff structure (for example, demand charges, energy charges and customer charges) strongly influence the optimized flexibility performance characteristics of each system (Fig. 3g–i). For example, the summer weekday tariff structure in Santa Barbara heavily incentivizes load shedding and peak power shaving from 16:00 to 21:00, as shown in Fig. 3a–c. However, in Houston, the time invariant (that is, flat) energy charge disincentivizes deviation from the optimally efficient baseline and results in high round-trip efficiencies and low energy and power capacities.

Operationalizing energy flexibility cost metrics

Combining energy flexibility performance metrics with individual facility and electricity market attributes ultimately informs the net benefits of flexible load operation for water treatment asset managers. The financial benefits of shifting load to avoid high energy and demand charges must be weighed against the additional costs of deploying native asset flexibility. Costs may stem from an increase the total electricity consumed, additional operating and maintenance costs at the facility or modest capital upgrades to sensing and control systems required for flexible load operation. These costs are accounted for within the proposed energy flexibility cost metrics, as discussed in assumptions and modelling approach.

We explore the financial implications of flexible operation across water assets as a function of net capital upgrade costs, facility age and tariff structure (Fig. 4). The LVOF of each water asset is a function of the capital upgrade cost and facility lifetime after upgrades (Fig. 4a–d). The LVOF for core water assets is positive above the NPV break-even line and financially preferable to the purchase and installation of a 1-LHE battery system (that is, battery sized to provide power to the facility for 1 h of average baseline facility operations) in cases above the 'battery' line. To generalize this case study, we then explore the sensitivity of the native LVOF to different tariff structures assuming a net-zero cost of upgrade and compare this to the value of a 1-LHE on-site battery (Fig. 4e). Finally, we provide the NPV of three diverse scenarios in tabular form (Fig. 4f).

Under the Santa Barbara Clean Energy tariff case, all water assets have a positive LVOF when the net capital upgrade costs are zero (Fig. 4a–d). Assets with more flexible loads (that is, high energy capacity) have higher LVOFs because they can shift more load outside peak electricity price periods. The intermittently operating desalination system with curtailment has both the highest normalized energy capacity (Fig. 3d) and the highest LVOF[27] (Fig. 4b).

The LVOF is sensitive to capital upgrade costs and remaining useful lifetime of the facility after upgrade (Fig. 4a–d). LVOF is most sensitive to capital upgrade costs at facilities with lower loads and/or more constrained energy discharge capacities, as highlighted in the water distribution case. As expected, upgrading facilities near the end of their useful life diminishes the LVOF. The LVOF can also exhibit small decreases at facilities with long expected lifespans as a result of the method discounting future benefits from flexible load operation over long time periods.

The break-even line (LVOF = 0) and the battery equivalency lines are highly specific to the water asset. For example, the advanced water treatment with curtailment case shows NPV-positive investments over a large range of lifetimes with capital upgrade costs in the millions of dollars, whereas the water distribution case is only NPV positive for very small capital upgrade costs. For the wastewater treatment and water distribution cases, installing a 1-LHE battery has a higher NPV than flexible load operation because these systems lack native flexibility (that is, tight time-varying production constraints) or have highly variable

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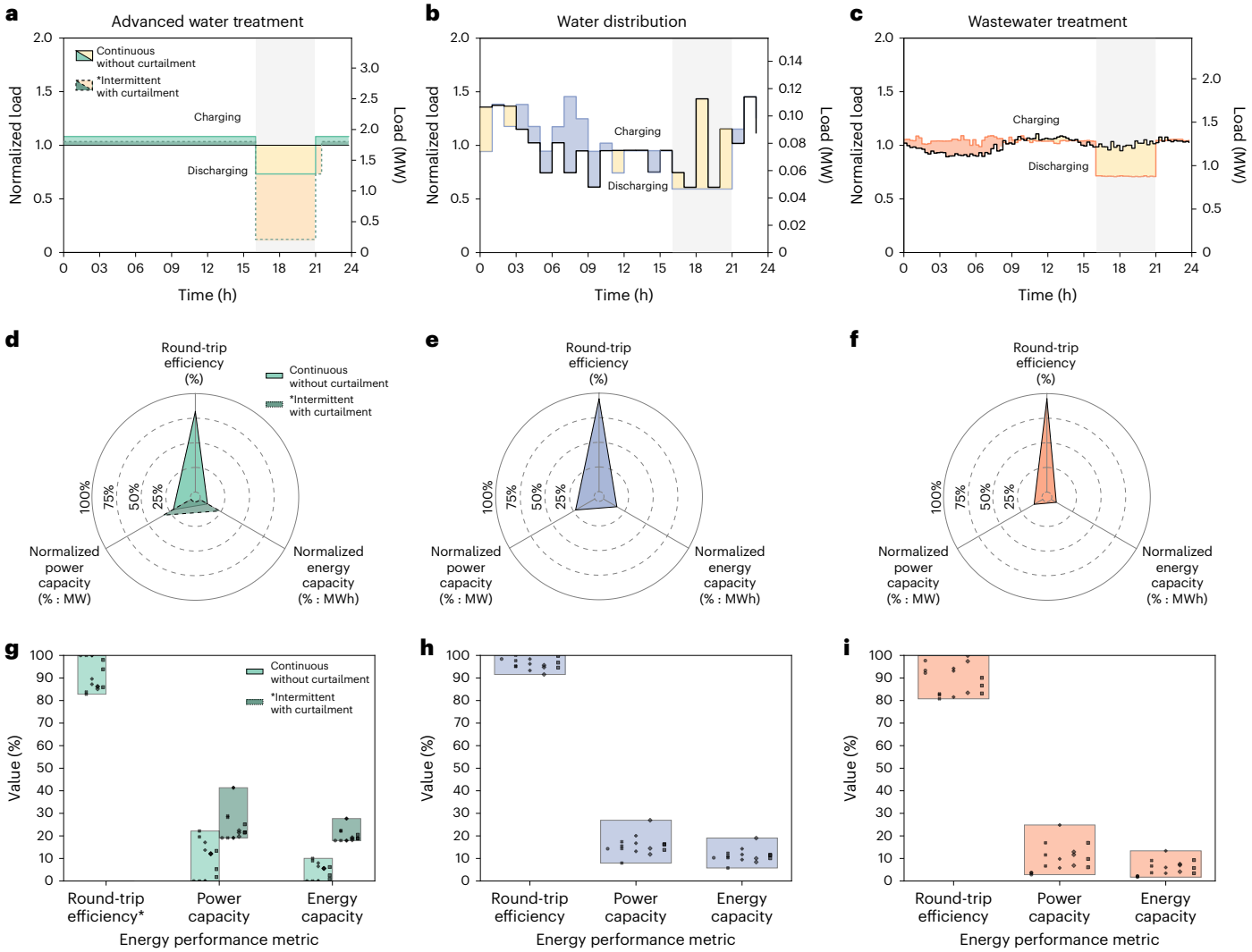

Fig. 3 | Energy performance metrics. a–c, Load profiles for an electricity cost-optimal flexible operating schema in advanced water treatment (**a**), water supply distribution (**b**) and wastewater treatment systems (**c**) are shown over a sample summer weekday. The baseline system operations are optimized based a constant electricity rate, whereas the flexible system operations are subject to a rate structure that resembles utility rates from Santa Barbara Clean Energy[36]. The peak electricity price hours (16:00–21:00) are highlighted in grey. **d–f**, The normalized, annual round-trip efficiency, energy capacity and power capacity associated with the Santa Barbara Clean Energy electricity tariff structure are shown. The energy capacity is normalized based on total baseline energy consumption. The power capacity is normalized based on the average baseline power consumption. **g–i**, Sensitivities on the energy performance metrics for each system are conducted with simulations using summer (July), spring (April) and winter (December) utility rate structures from Houston, New York, San Jose, Santa Barbara and Tampa[36]. These cities were selected to represent a diverse range of tariff structures and average electricity prices. The round-trip efficiency is not defined for the supply curtailment scenario as it does not complete a round trip. All cases of desalination supply curtailment assume the baseline equivalent of 5 h of lost water production, which may be representative of typical demand-response events.

baseline loads. Past work has also demonstrated synergistic benefits between flexible load operation and battery storage at wastewater treatment facilities, where coordinating battery charging with process operations can increase biogas utilization and further enhance facility flexibility[12]. By contrast, assets where the baseline operation is near steady state derive little added value from an on-site battery. Flexible load operation of advanced water treatment is notably more cost effective than steady state operation with battery storage for most cases.

The LVOF of an asset is also highly sensitive to the magnitude and timing of price variation within a tariff structure (Fig. 4e). Facilities operating under highly time variant tariff structures (that is, high peak–off-peak ratio; PG&E and Santa Barbara Clean Energy) are more strongly incentivized to operate flexibly. Even so, the timing of the tariff structure's peak and off-peak intervals must align with the timing of an asset's native flexibility to realize a high LVOF. None of the examined assets can fully shift their load out of the peak price period due to system constraints such as water treatment, water delivery or tank storage limitations (Fig. 3a–c).

LVOF is a valuable metric for unifying the comparison of savings from flexibility across and outside of the water sector; however if the NPV or ROI is too low, facilities may decide not to deploy native flexibility (Fig. 4f). For instance, the water distribution case highlighted in

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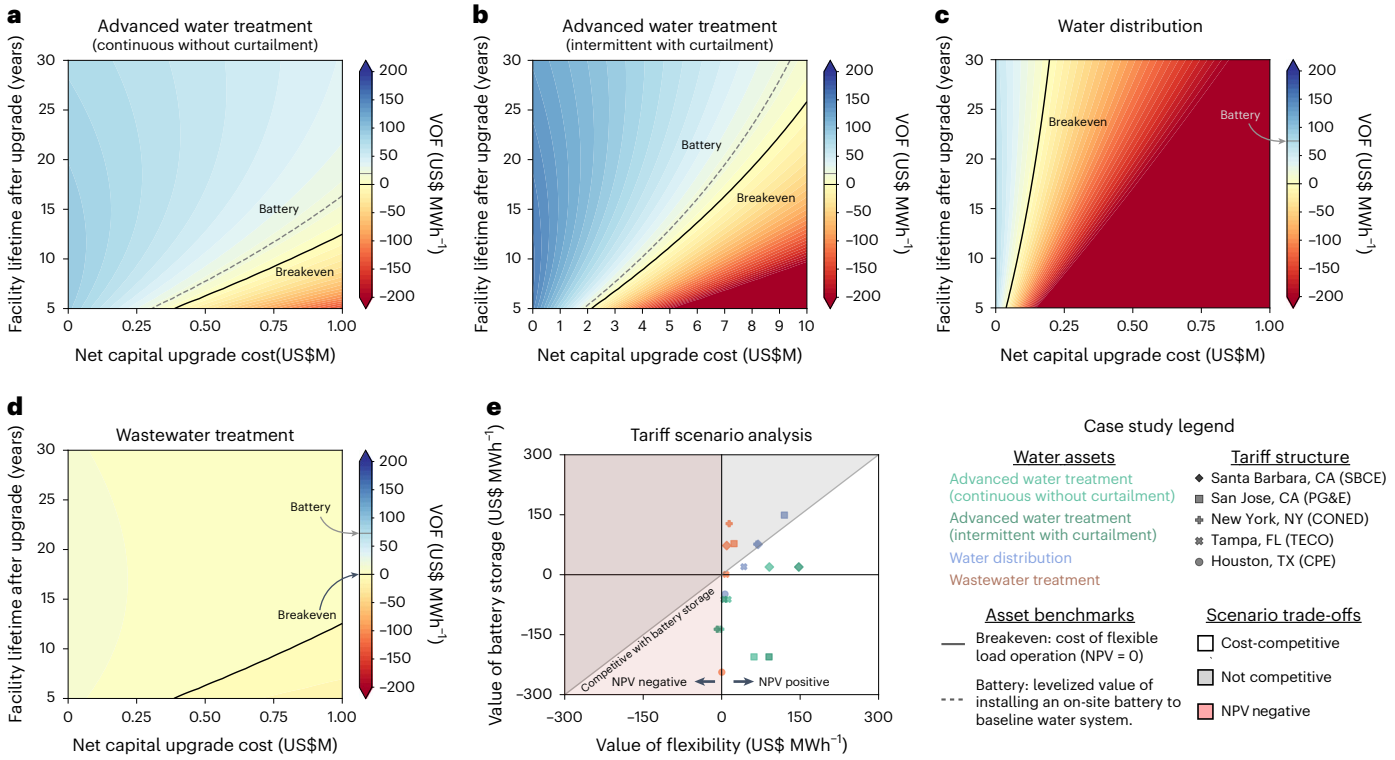

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Water utility	Energy provider	Symbol	Net present value (US$)	Energy capacity (MWh)	Levelized cost of flexibility (US$ MWh^{-1})	Value of battery storage (US$ MWh^{-1})
Advanced water treatment (intermittent with curtailment)	Santa Barbara Clean Energy	◆	1,553,000	17,000	91.1	19.3
Water distribution	Tampa Electric Company	✖	65,000	1,500	42.4	19.9
Wastewater treatment	Consolidated Edison	✚	2,375,000	165,800	14.3	127.8

Fig. 4 | Levelized value of flexibility. a–d, The LVOF is displayed for each of the systems described in Fig. 3, using the Santa Barbara Clean Energy electricity tariff structure. The net upgrade costs represent the costs associated with facilities upgrades that enable operational flexibility (for example, a new control system or software update) and are approximated by ranges in literature[49,50]. The range of capital upgrade cost is extended to represent unit process changes that may be required for an advanced water treatment to restart quickly after a shutdown event. The facility lifetime represents the number of years remaining in the facility's economic lifetime or the years over which the capital investment upgrades will be amortized. The solid isoline represents the equilibrium space for NPV-positive upgrades. The dashed isoline represents the value of installing a 1-load-hour-equivalent lithium-ion battery at a capital cost of US$450 kWh^{-1} (ref. 48). In the water distribution and wastewater treatment cases (**c**,**d**), the battery contour line is not present on the chart but referenced within the colour scale and fully labelled in **d**. **e**,**f**, Consider a sensitivity of LVOF to tariff rate structures and asset classes assuming a net capital upgrade cost of US$0 (ref. 36). In these cases where native flexibility is considered (capital upgrade cost = $0), return on investment is undefined. All cases of desalination supply curtailment assume the baseline equivalent of 5 h of lost water production, which is representative of typical demand-response events across tariffs. Advanced water treatment under the Houston tariff structure is not shown due to negligible differences between the flexible and baseline operating schedules.

Fig. 4f has a competitive LVOF, but since the asset has a much smaller energy load, the magnitude of lifetime savings is low. In contrast, the NPV of the wastewater treatment case is the largest across the examined cases but yields the lowest LVOF due to the large absolute energy capacity. Whereas native flexibility has high NPV in this case, the LVOF metric allows the operator to determine that installing a battery would be a much more valuable investment.

Finally, the existing asset configurations modelled above are often constrained in their flexibility by one or more components (for example, inadequate capacity, no intermediate water storage, static pumps). To alleviate these facility-level constraints, capital improvement plans might include investments to enhance energy flexibility. Facility upgrades that unlock new energy or power capacity will often have LVOF values that are not readily extrapolated from the above plots. In some cases, such as those where there is a great deal of native flexibility in the system, there are diminishing marginal returns from additional flexibility. In other cases, enhanced flexibility is expected to be super linear because of newly imparted ability to avoid high demand charges. Water system operators would benefit from additional scholarship exploring the specific constraints on flexibility at facilities and developing supply curves for energy flexibility upgrades based on technology options and optimized operations. An understanding of a facility's capacity-cost relationship for energy flexibility, which will vary change seasonally with capacity requirements, is key in determining the frequency at which flexible loads may be used and their bulk effects on the grid over longer timescales.

Outlook

Realizing the grid benefits of water asset flexibility requires a common framework for valuing key metrics across diverse electricity and water infrastructure systems. The proposed framework characterizes the value of flexible load energy services to the grid and the value of those services to water facility operators seeking to minimize growing

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a Case study electricity tariff structures

Water utility	Electricity provider	Months	Peak hours	Symbol
Santa Barbara, CA	Santa Barbara Clean Energy	June–Sept	Mon–Fri 16:00–21:00	◆
		Oct–May	Mon–Fri 16:00–21:00	◆
San Jose, CA	Pacific Gas & Electric	June–Sept	Mon–Sun 16:00–21:00	■
		Oct–May	Mon–Sun 16:00–21:00	■
Tampa, FL	Tampa Electric Company	Apr–Oct	Mon–Fri 12:00–21:00	✖
		Nov–Mar	Mon–Fri 06:00–10:00/18:00–22:00	✖
New York, NY	Consolidated Edison	June–Sept	Mon–Fri 08:00–18:00	✚
		Oct–May	Mon–Fri 08:00–22:00	✚
Houston, TX	CenterPoint Energy	Jan–Dec	NA	●

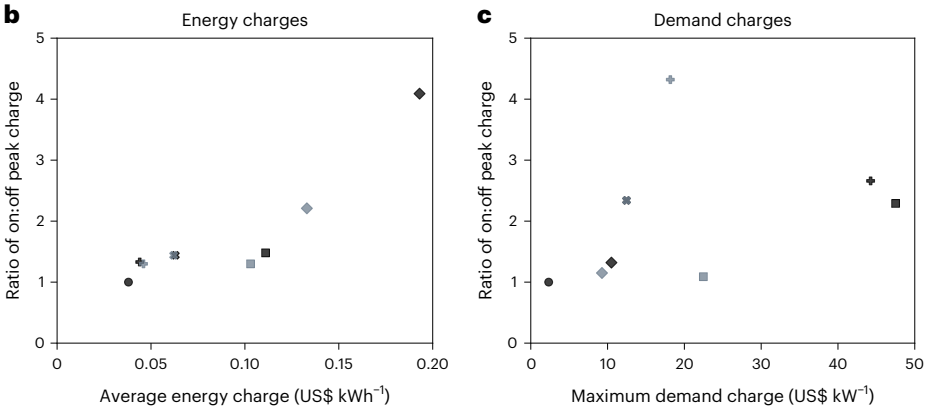

Fig. 5 | Electricity tariff scenario case studies. a, Summarizes the electricity tariffs used in scenario analysis of our case study water facilities. Selected water utilities each purchase power from a single electricity provider. The tariff structure and rates (energy and demand charges) differ across electric power utilities and, in some cases, vary both seasonally and diurnally. Months refer to the times of year when the specified charges apply, whereas peak hours refer to the days and times of peak energy and demand prices. NA, not applicable (that is, the Houston tariff structure does not contain any temporal variation). **b**,**c**, The corresponding symbol for each tariff structure is plotted to assist in comparing the average (energy) or maximum (demand) charge with the variation of this charge within a representative day. The energy charge is the marginal rate of energy use at all time steps. The peak demand charge is the rate associated with the maximum 15-minute average power consumption during peak hours. Santa Barbara and San Jose tariffs include multiple peak and off-peak price tiers. Full tariffs can be found in Chapin et al.[36].

electricity costs. Grid operators can use this framework to evaluate the magnitude and dispatchability of distributed energy resources (DER) across diverse water asset classes, thus enabling direct comparisons to DERs from other sectors. They might also use these metrics to help modify and price energy service programmes in an evolving grid. These programmes may involve sector-specific tariff structures with real-time pricing or versions of incentive-based demand response to further motivate load following behaviour.

Water utility operators can use this framework to inform financial decisions about flexible load operation of diverse water assets, prioritize capital improvements to enhance flexibility within and across facilities, and evaluate the benefits of non-native flexibility enhancements such as on-site battery installations. Water facility designers might also use this framework with site-specific costing data, to inform the design of water assets with high-value native flexibility. Future work should employ this framework for a larger number of facilities, tariff structures and ancillary service markets to assess the projected future opportunity for individual plants to profitably provide energy services. Future work should employ this framework for a larger number of facilities, tariff structures and ancillary service markets to assess the projected future opportunity for individual plants to profitably provide energy services.

Whereas valuing the benefits of water asset flexibility is critical to dispatch of this flexibility, there are several additional barriers to flexible load operation of water assets. Past work has highlighted risk aversion in critical infrastructure operation, potential facility-specific implications for water quality associated with non-standard operation, long payback periods, investment cycles with high uncertainty, a shortage of labour certified to operate these systems, tension with static energy efficiency improvements, operating and maintenance costs associated with flexible operations and concerns about energy market transparency[13,38].

Regardless of these concerns, an objective set of metrics for characterizing the performance and value of flexibility can help academics, policymakers and system operators to de-risk and optimize future flexible infrastructure operations. Similar formulations of these metrics may be extended beyond the water sector to evaluate and optimize the general case of an industrial process with arbitrary forms of energy inputs (for example, grid electricity, heat and so on) and where decision-making is governed by lifetime economics.

Methods

Modelling approach

We operationalize the metrics defined this work using simplified energy models of a seawater reverse osmosis system, a water distribution network and a secondary wastewater treatment facility with biogas cogeneration. Conventional water treatment systems and water transmission systems are not considered here, as these systems generally represent a small portion of the energy consumption of a water utility and are limited in their total load flexibility[39]. One exception to this case includes the California State Water project, which is the largest single electricity consumer in California, as of 2023[40].

Case study models

The seawater reverse osmosis system model uses the WaterTAP case study Seawater Reverse Osmosis treatment train (Fig. 1a) with a baseload model energy intensity for each process unit[41,42]. The continuously flexible system operates with a variable recovery ratio, subject to water production requirements. The intermittent system shuts down the energy-intensive reverse osmosis process during the peak price hours but does not curtail intake pump operation, pretreatment or other baseload energy demands at the facility. The intermittent case also proportionally relaxes the water production requirement (that is, 5 of 24 hours are non-producing, or a 21% curtailment). The water distribution model (Fig. 1b) uses a modified skeletonized benchmark system[29]. Pump status (on and off) is optimized over a 24-h period using the methods described previously[39] assuming a maximum of 10 switches per day per pump to limit component degredation. The wastewater treatment load model (Fig. 1c) is based on methods outlined previously[12]. The flexible system is optimized with the storage of wastewater and biogas storage.

In all cases, the optimized flexible system aims to minimize the electricity cost of the system over the time horizon, subject to the case study tariff structure, process dynamics and production flow-rate constraints. In the desalination case, the system is constrained to meet a daily water production target (continuous without curtailment), or a reduced water production target (intermittent with curtailment). In the water distribution case, water delivery constraints are applied to each consumer node of the network on hourly intervals. For wastewater modelling case, the production constraint is generated via stochastic realizations of influent data that vary on 15-minute intervals. The optimization models are solved using Pyomo (advanced water treatment and water distribution) and CVXPY (wastewater treatment)[43–47].

Electricity tariffs

We perform scenario analysis (Fig. 5) to investigate the effect of electricity tariff structure on the economic viability of flexible load operation. For simplicity, we assume tariffs associated with five wastewater utilities are representative of tariffs for other local water assets[36]. The tariff structures represent a broad range of nationally representative cases that vary in electricity price and time-of-use structures.

Model assumptions

These metrics require implicit assumptions for averaging and aligning short, simulated time horizons (for example, one day, week, year) with time frames for financial decisions that are made annually or once in a facility's economic lifetime. All simulations in this paper calculate the expected costs of flexibility over a one-year period that is assumed to be representative of the 30-year facility lifetime. Simulations optimize flexible operations for the electricity cost with a series of representative 1-day or 1-month time horizons. Unless otherwise noted, all cases assume that flexible operation of facilities imposes additional OM costs of US$10,000 per annum. As described above, this OM cost term captures both accelerated component degradation and additional maintenance of capital assets.

Optimized flexible load operation is compared to the baseline scenario and a case in which a physical battery is installed at the water facility. The battery is modelled with a round-trip efficiency of 86%, an energy capacity equivalent to 1-load-hour of baseline operation and a power capacity associated with the maximum power draw from the baseline facility. The capital cost of the battery is modelled at US$450 kW^{-1} for 4-hour lithium-ion systems, reflecting the performance and cost parameters of commercially available battery systems[48].

Data availability

Data are publicly available via GitHub at https://github.com/we3lab/valuing-flexibility-from-water.

Code availability

Code for all figures with data is publicly available via GitHub at https://github.com/we3lab/valuing-flexibility-from-water. Additionally, we provide an interactive code notebook with download instructions to provide more in-depth visualizations for specific cases.

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Author contributions

A.K.R.: conceptualization, methodology, software, validation, formal analysis, investigation, writing (all stages), visualization, project administration, funding acquisition. J.B.: conceptualization, methodology, software, formal analysis, writing (review and editing). E.M.: methodology, software, formal analysis, writing (review and editing). F.T.C.: methodology, software, data curation, writing (review and editing). M.S.M.: conceptualization, resources, supervision, funding acquisition.

Competing interests

The authors declare no competing interests.

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